Stratum 4D
by Miqyas Labs
PLAN | SIMULATE | ANALYSE | DELIVER

User Manual

Release 0.10.1 · Planning, 4D simulation, dashboards and reports

Written in ASD-STE100 Simplified Technical English.
Stratum 4D by Miqyas Labs · developed by Junaid Quamar · linkedin.com/in/junaidquamar
October 2026. The sample projects in this manual are fictional.

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1 Introduction

This chapter tells you what Stratum 4D does, which layouts it has, and how to use this manual.

What Stratum 4D is

Stratum 4D is a planning and 4D simulation tool for construction projects. It reads a schedule from Primavera P6 or Microsoft Project. It can also read a 3D building model in the IFC format.

You link the model elements to the activities of the schedule. Stratum then shows the construction sequence in 3D, day by day. You can calculate the schedule, record progress, compare against a baseline and make reports.

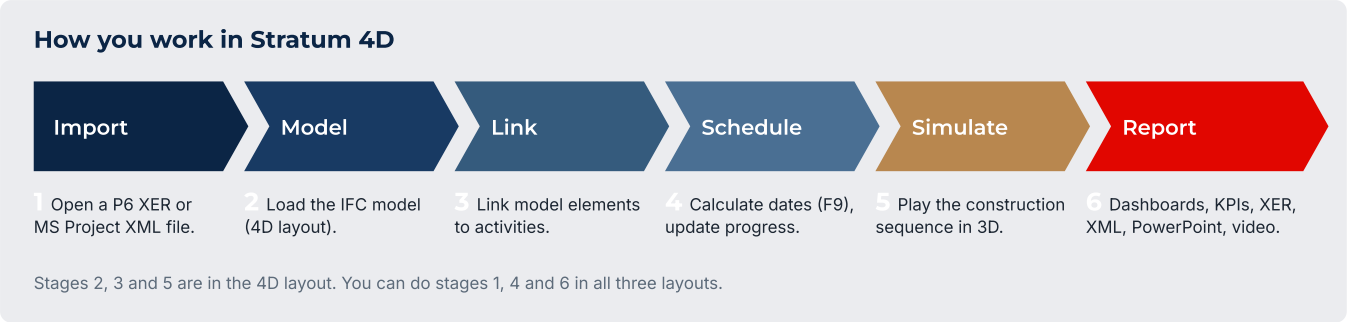


Figure 1-1. The six stages of work in Stratum 4D.

The three layouts

Stratum 4D has three layouts. All three layouts use the same calculation engine and the same project data. When you change the layout, the open project stays open.

- The **P6** layout looks and works like Primavera P6 Professional.
- The **Project** layout looks and works like Microsoft Project.
- The **4D** layout shows the 3D model, the Gantt chart and the playback timeline together.

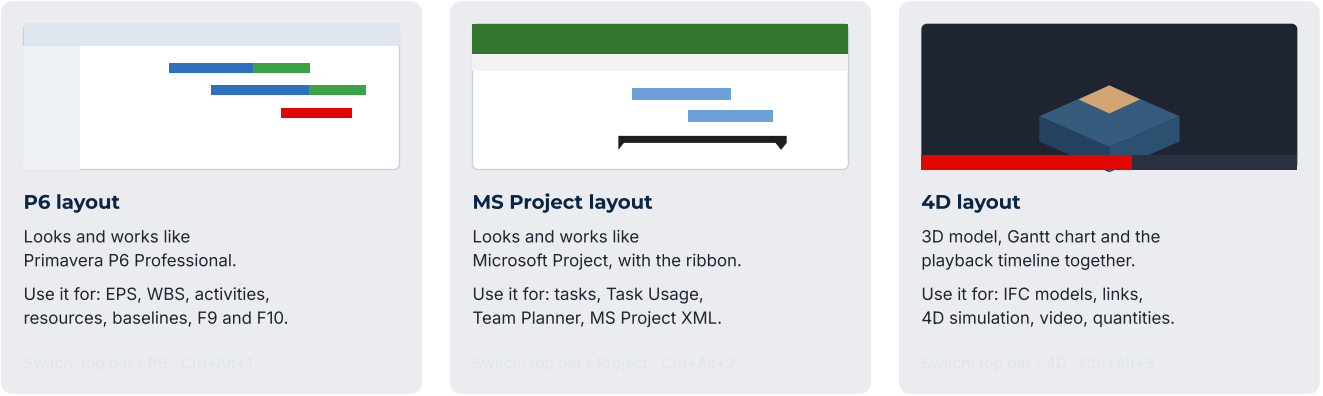


Figure 1-2. The three layouts and what each layout is for.

To change the layout

1. On the top bar, click **P6**, **Project** or **4D**.
   Result: The new layout opens with the same project.

2. Alternatively, push `Ctrl` + `Alt` + `1`, `2` or `3`.

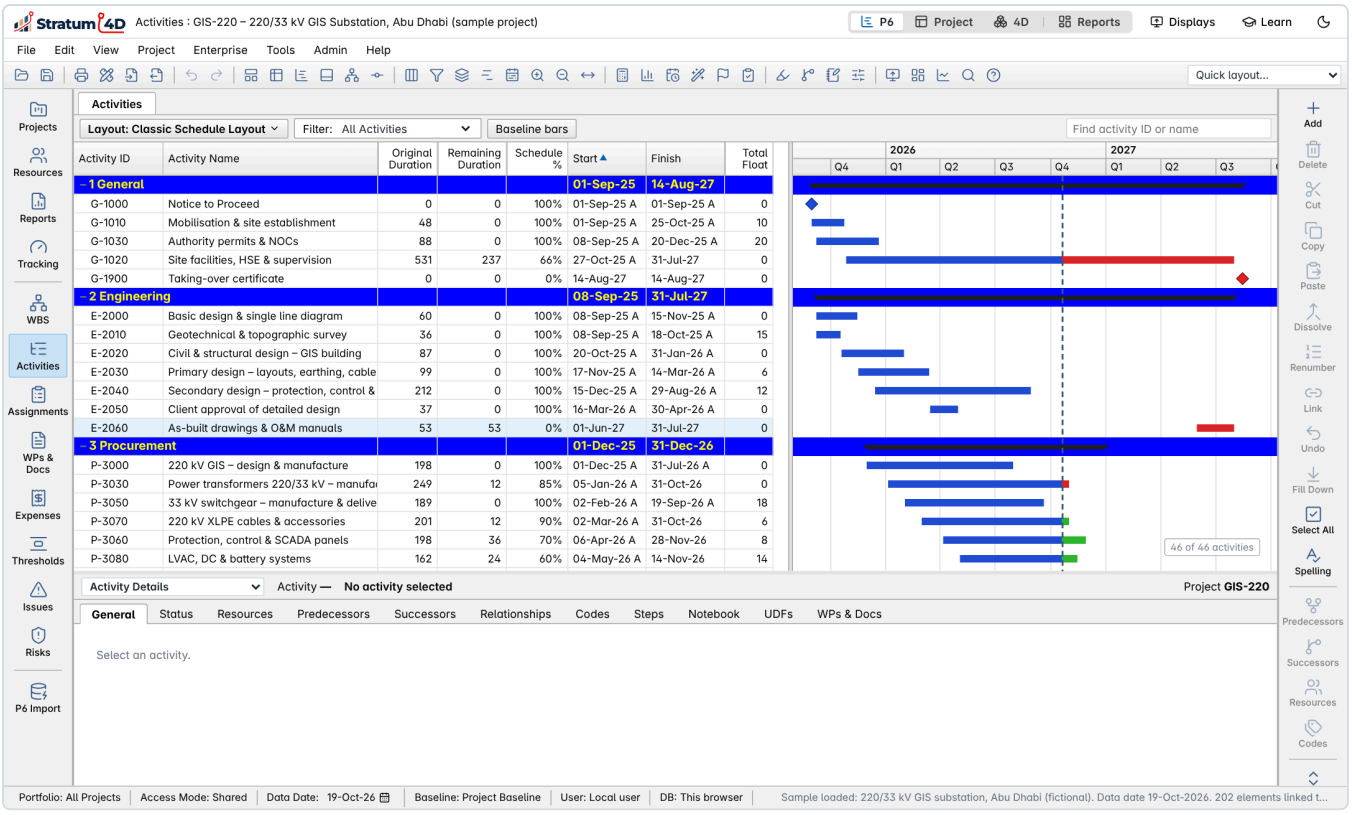


Figure 1-3. The P6 layout with the 220 kV GIS substation sample project.

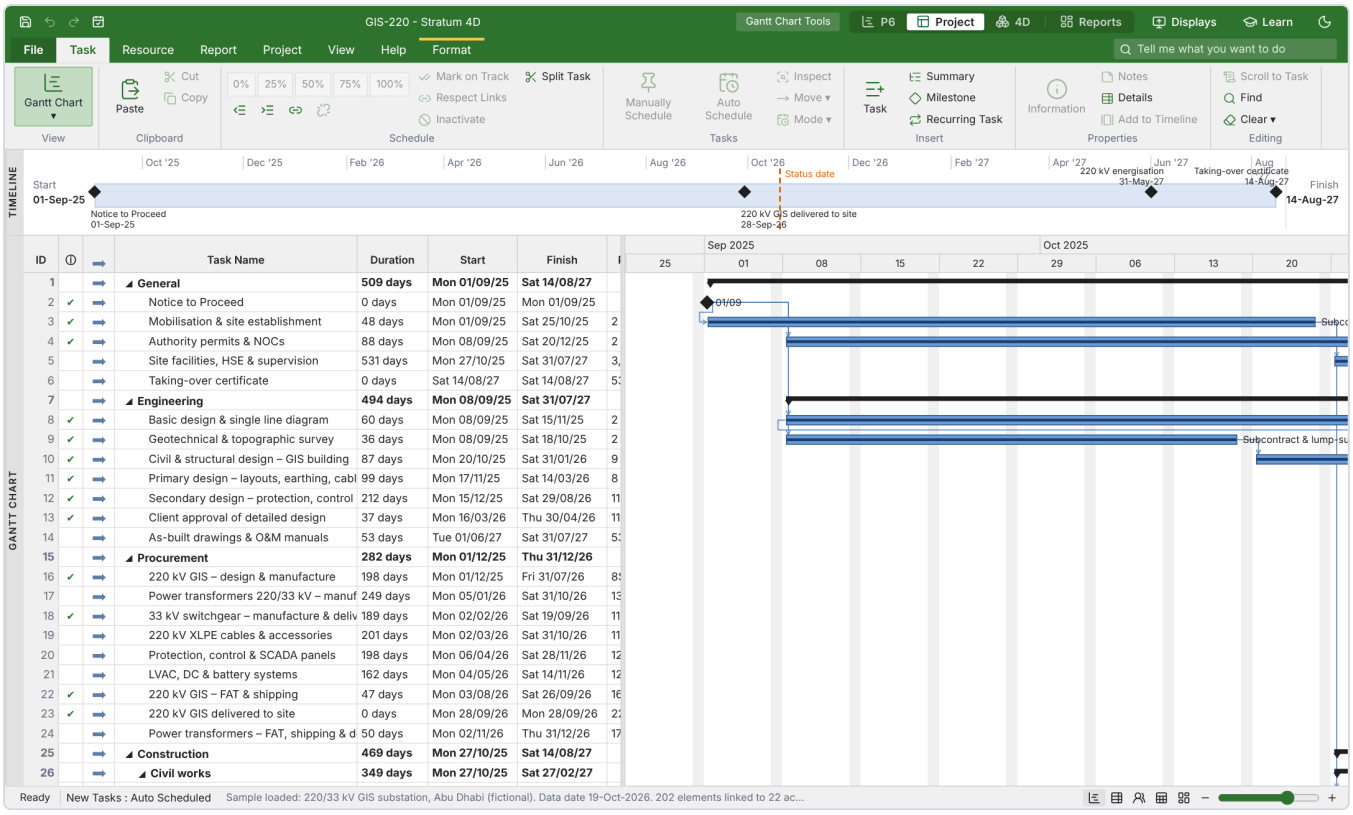


Figure 1-4. The Project layout (Microsoft Project style) with the same project.

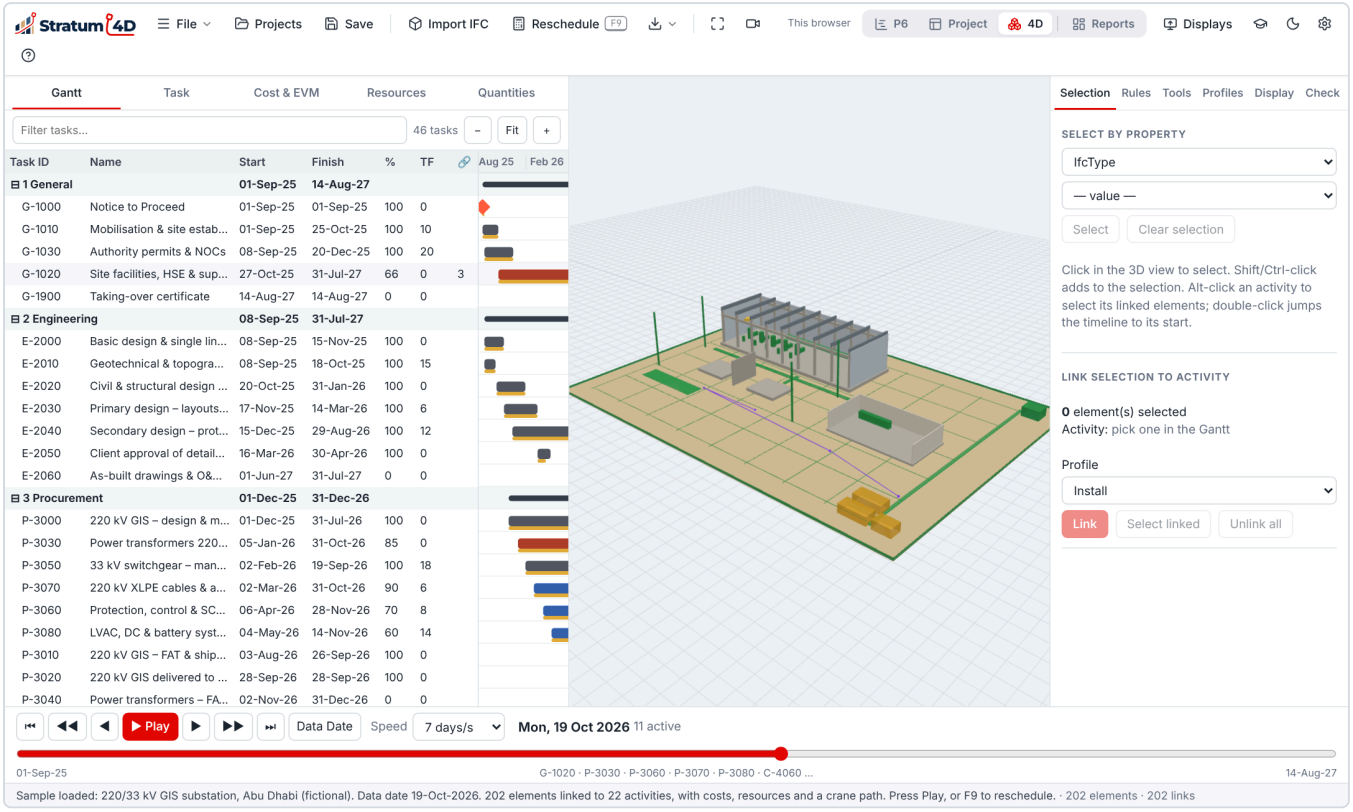


Figure 1-5. The 4D layout: Gantt chart on the left, 3D model in the middle, link tools on the right and the timeline at the bottom.

The sample projects

Stratum 4D has two sample projects. You can use them to learn the software without your own files. The two sample projects are fictional.

Sample project	Contents
220 kV GIS substation (UAE)	46 activities on a UAE 6-day calendar, costs in AED, resources, a baseline and a 3D model with 202 elements.
Small RC building	A small concrete building with 14 activities and a 3D model. Use it for a fast first test.

This manual uses the 220 kV GIS substation project for most examples.

How to use this manual

This manual is written in ASD-STE100 Simplified Technical English. Each chapter starts with its purpose. Then it gives a description and the procedures.

- A procedure is a numbered list of steps. Do the steps in the given order.
- A line that starts with "Result" tells you what you see after the step.
- Names of buttons, menus and fields are in **bold blue**.
- Keys are shown as `Ctrl` + `S`. Push the keys at the same time.

> **NOTE**
> A NOTE gives information that helps you.

> **CAUTION**
> A CAUTION tells you about a risk of loss of data.

The words in the layouts

Each layout uses the words of the software that it copies. The table gives the words for the same item in each layout.

Item	P6 layout	Project layout	4D layout
One item of work	Activity	Task	Task
Its identifier	Activity ID	ID (row number)	Task ID
Date of the last progress update	Data date	Status date	Data date
Calculate the dates	Schedule (F9)	Calculate Project (F9)	Reschedule (F9)
Time that work can slip	Total float	Total slack	TF
Group of work	WBS	Summary task	WBS

Related chapters: 2 Installation, 3 First start, 4 Projects and files.

2 Installation

This chapter tells you how to install Stratum 4D on Windows, macOS, Linux and Android, and how to use it in a web browser.

Versions of Stratum 4D

Stratum 4D is available as a desktop application, an Android application and a web application. All versions have the same layouts and the same calculations.

The desktop application can connect to Primavera P6 and can send reports by email. The web application and the Android application cannot connect to P6.

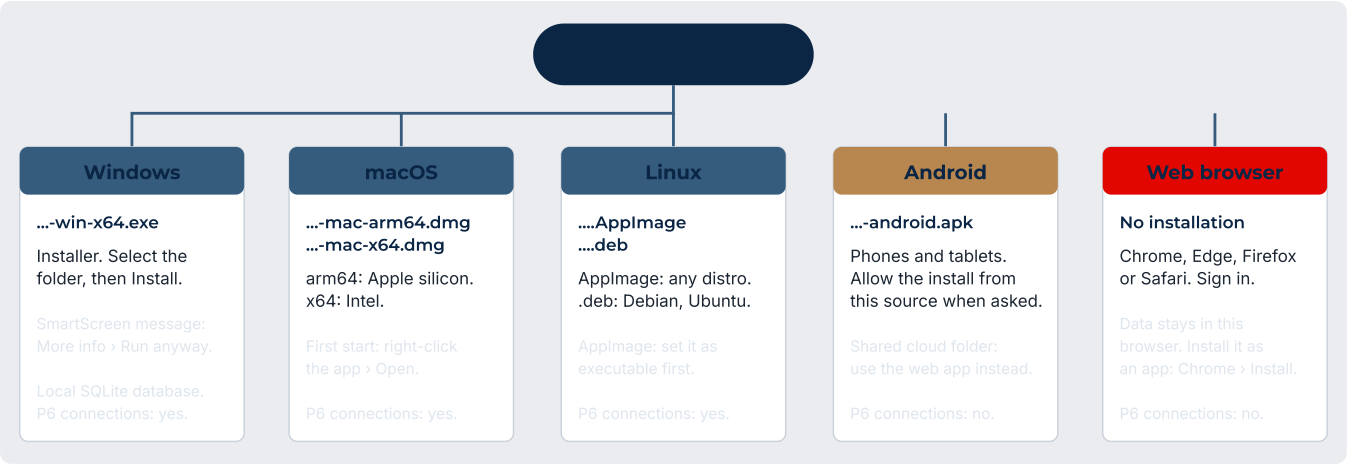


Figure 2-1. Select the correct file for your device.

Get the installation files

The installation files are on the releases page of Stratum 4D on GitHub. Each release has one file for each type of device.

1. Open the releases page: `github.com/xunaidlnt-sudo/stratum-4d/releases`.
2. Find the latest release at the top of the page.
3. Under **Assets**, click the file for your device.
   Result: The file goes into the Downloads folder.

Install on Windows

Use the file that ends with `-win-x64.exe`. Windows 10 and Windows 11 (64-bit) are supported.

> **NOTE**
> The installer does not have a code signature yet. Thus Windows SmartScreen can show the message "Windows protected your PC".

1. Double-click the file `Stratum-4D-<version>-win-x64.exe`.
2. If SmartScreen shows a message, click **More info**.
3. Click **Run anyway**.
4. Select the folder for the installation, or keep the default folder.
5. Click **Install**.
   Result: The installer adds Stratum 4D to the Start menu and to the desktop.
6. Click **Finish**.
   Result: Stratum 4D starts.

After the installation, Windows opens `.xer` and `.ifc` files in Stratum 4D when you double-click them.

Install on macOS

There are two files for macOS. Use `-mac-arm64.dmg` on a Mac with Apple silicon (M1 or later). Use `-mac-x64.dmg` on a Mac with an Intel processor.

1. Double-click the `.dmg` file.
   Result: A window opens with the Stratum 4D icon.
2. Drag the Stratum 4D icon to the **Applications** folder.
3. Open the **Applications** folder.
4. Right-click **Stratum 4D** and click **Open**.
5. In the message, click **Open** again.
   Result: Stratum 4D starts. Next time, you can open it as usual.

Install on Linux

There are two files for Linux. The `.AppImage` file operates on most distributions. The `.deb` file is for Debian and Ubuntu.

To use the AppImage file

1. Open a terminal in the Downloads folder.

2. Type `chmod +x Stratum-4D-*.AppImage` and push `Enter`.
3. Double-click the AppImage file.
   Result: Stratum 4D starts.

To install the .deb file

1. Open a terminal in the Downloads folder.
2. Type `sudo apt install ./Stratum-4D-*.deb` and push `Enter`.
3. Start Stratum 4D from the applications menu.

Install on Android

Use the file that ends with `-android.apk`. The application is for phones and tablets with Android 8 or later.

1. On the phone or tablet, open the releases page and download the `.apk` file.
2. Open the file from the download notification.
3. If Android asks for permission to install from this source, give the permission.
4. Tap **Install**.
5. Tap **Open**.
   Result: Stratum 4D starts.

> **NOTE**
> To share a cloud folder (Google Drive or OneDrive) with a team on a tablet, use the web application. Chapter 4 tells you about shared folders.

Use the web application

The web application operates in Chrome, Edge, Firefox and Safari. It is not necessary to install it. The web application keeps your projects in the storage of the browser on that computer.

1. Open the address of the web application that you got from Miqyas Labs.
2. Type your user name and password.
3. Click **Sign in**.
   Result: Stratum 4D opens. At the first start, the setup steps of chapter 3 open.

To install the web application as an application

1. Open the web application in Chrome or Edge.
2. In the address bar, click the install icon.
3. Click **Install**.
   Result: Stratum 4D opens in its own window, with an icon on the desktop.

> **CAUTION**
> If you clear the data of the browser, the projects in the browser storage are deleted. Save a project file or use a shared folder to keep a copy (chapter 4).

Update to a new release

To update the desktop application, install the new release over the old release. Your projects and settings stay. The web application updates when you open it again.

Remove Stratum 4D

- Windows: open **Settings › Apps**, select **Stratum 4D** and click **Uninstall**.
- macOS: drag **Stratum 4D** from the Applications folder to the Bin.
- Linux: delete the AppImage file, or type `sudo apt remove stratum-4d`.
- Android: touch and hold the icon, then tap **Uninstall**.

> **CAUTION**
> The projects in the database stay on the computer after you remove the application. Delete them only if you have a copy.

Related chapters: 3 First start, 4 Projects and files.

3 First start

This chapter tells you what to do when you start Stratum 4D for the first time.

The setup steps

At the first start, Stratum 4D asks two questions: which layout you want, and where to keep the projects. You can change both answers later in **Settings**.



Figure 3-1. The steps at the first start.

Select the layout

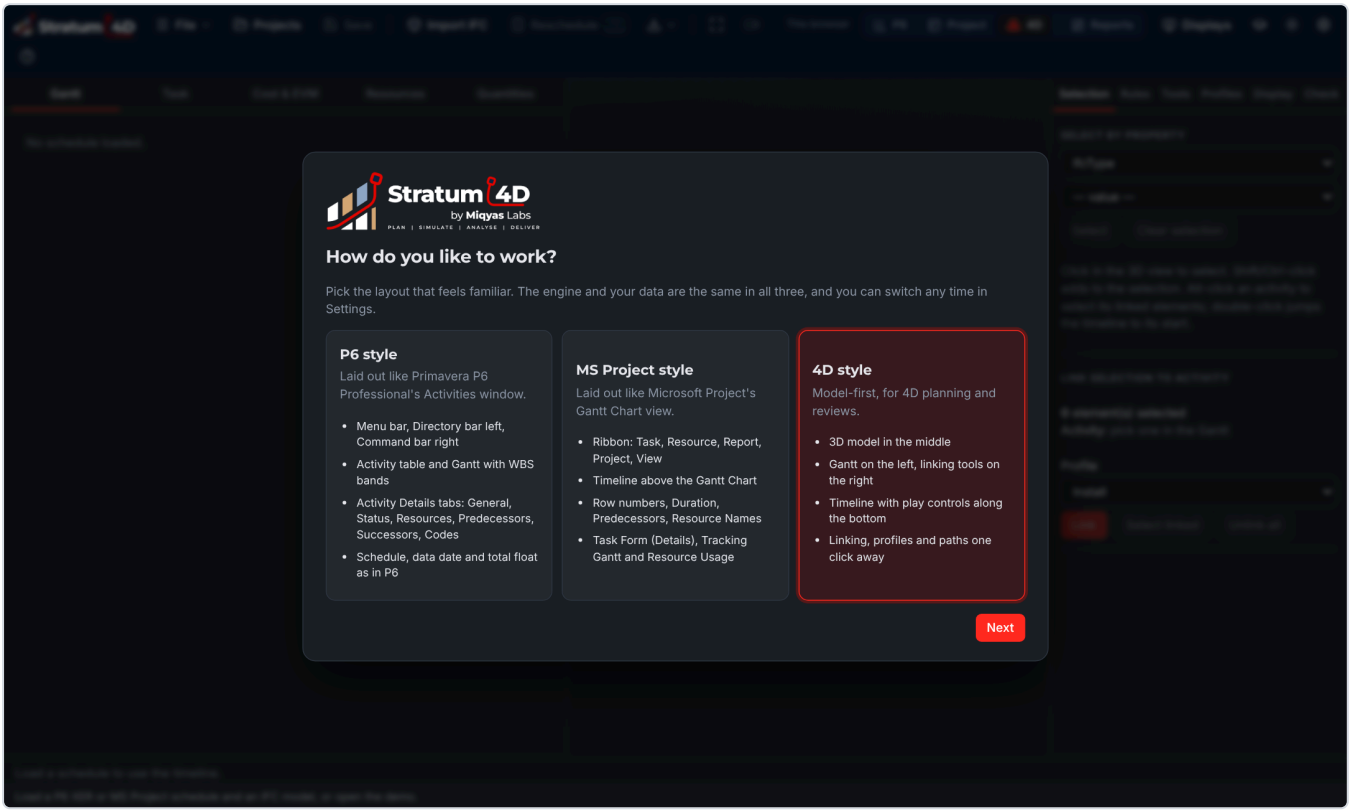


Figure 3-2. Select the layout that you know best.

1. Click **P6 style**, **MS Project style** or **4D style**.
2. Click **Next**.

> **NOTE**
> If you work with Primavera P6 every day, select **P6 style**. If you make 4D simulations, select **4D style**.

Select the storage

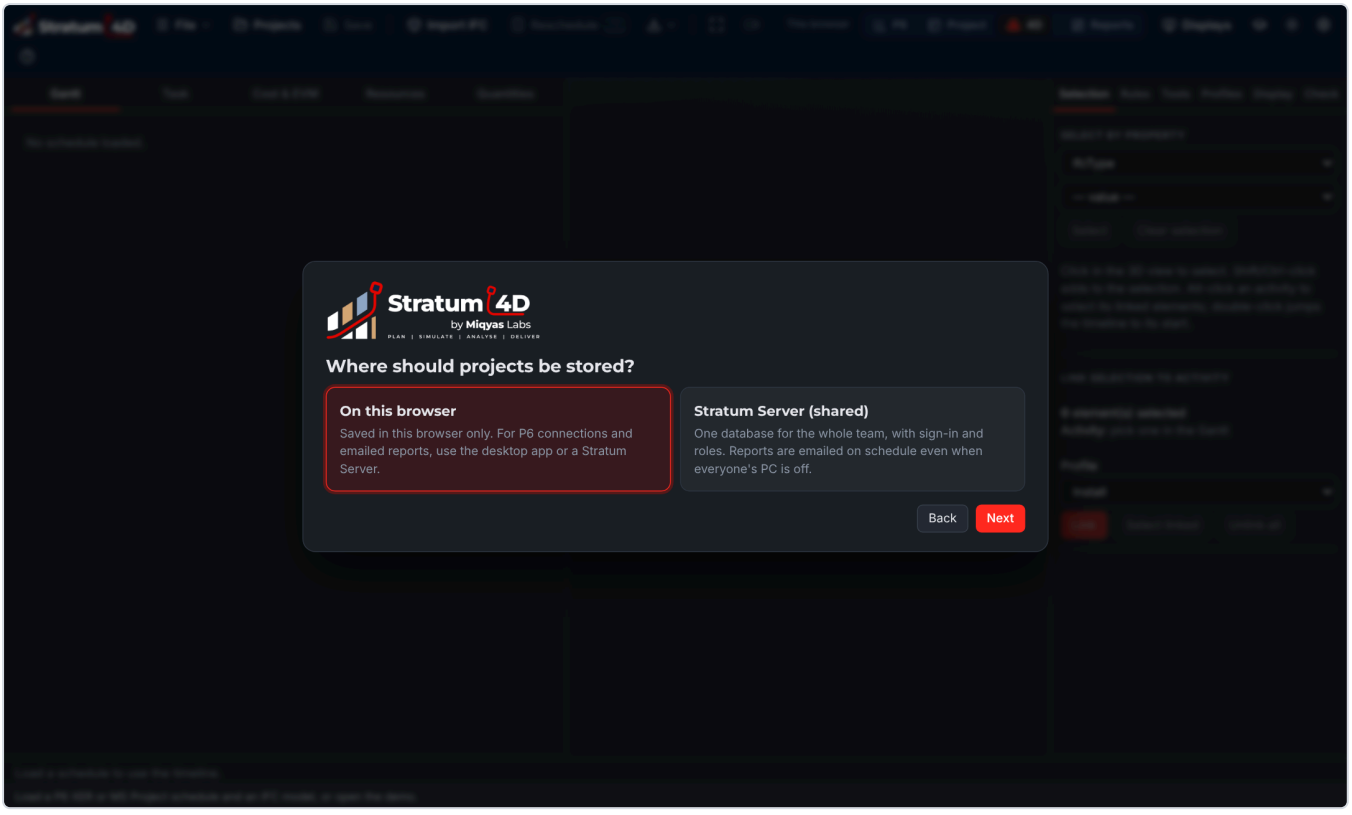


Figure 3-3. Select where Stratum 4D keeps the projects.

Storage	Use it when
On this computer (desktop) or On this browser (web)	You work alone. The projects stay on this computer.
Stratum Server (shared)	Your team has a Stratum Server. All users work on the same projects, with sign-in and roles.

To keep the projects on this computer

1. Click **On this computer** (or **On this browser**).
2. Click **Next**.

To connect to a Stratum Server

1. Click **Stratum Server (shared)**.

2. Type the address of the server that you got from your administrator.
3. Click **Check**.
   Result: Stratum 4D finds the server.
4. If the server is new, type a name, an email address and a password for the first administrator. Then click **Create admin and connect**.
5. If the server has users, type your email address and password. Then click **Sign in**.
   Result: The step **Ready** opens.

Optional connections

The last step, **Ready**, lets you connect to Primavera P6 and set up email for reports. These two buttons show only in the desktop application and with a Stratum Server. You can skip this step. Chapter 4 and the chapter about P6 connections tell you how to do this later.

1. To connect to P6 now, click **Connect to P6...**.
2. To set up email now, click **Set up email...**.
3. To skip, click **Start**.
   Result: The workspace opens.

The empty workspace

When no project is open, the workspace shows a short message. From it, you can open a sample project or start the training.

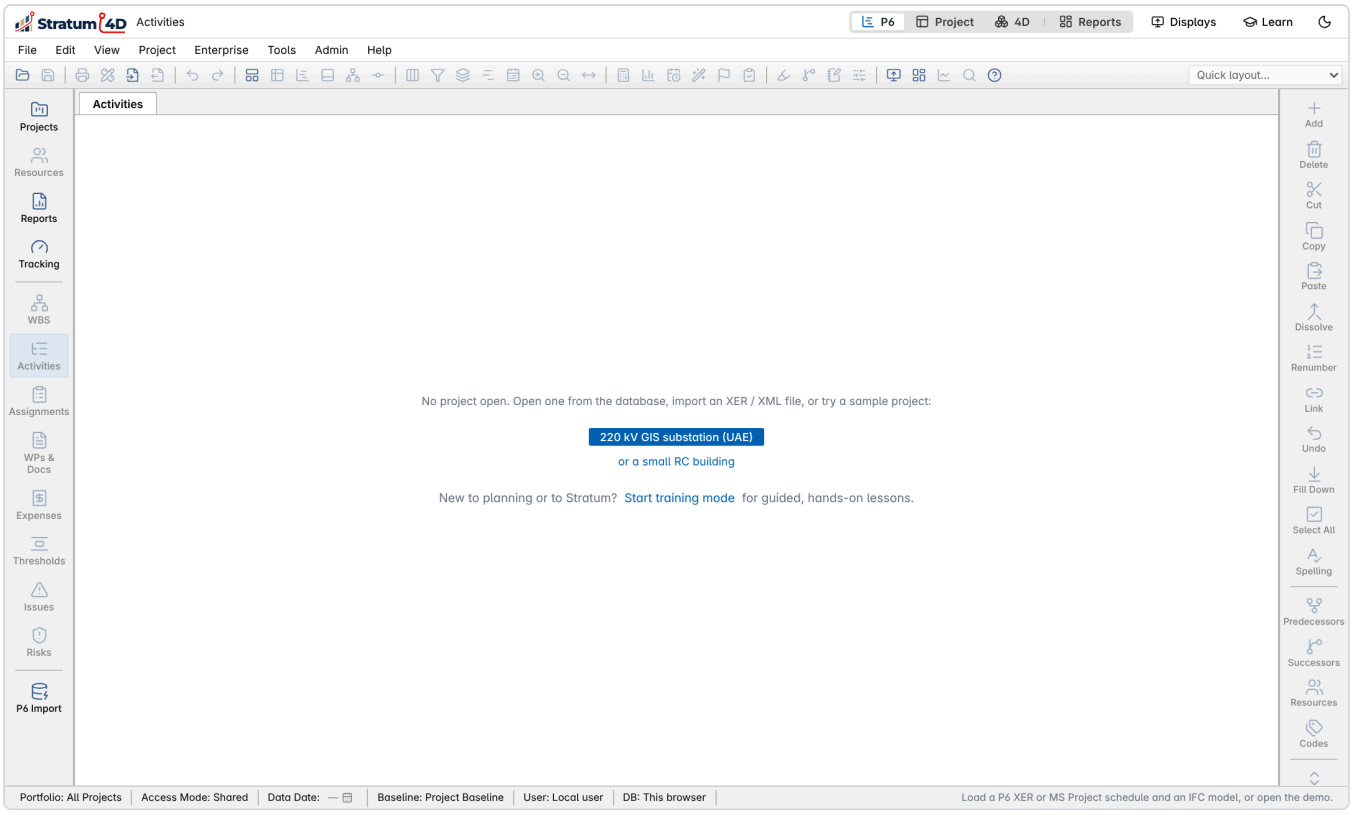


Figure 3-4. The empty workspace in the P6 layout.

To open a sample project

1. Click **220 kV GIS substation (UAE)**.
   Result: The sample project opens in the layout that you selected.

To start the training

1. Click **Start training mode**, or click **Learn** on the top bar.
2. Select a lesson or a course.
   Result: The lesson shows you each step in the software.

Change the settings later

1. On the top bar, click the gear icon (**Settings**).
2. Select a tab: **Interface**, **Data & sign-in**, **Primavera P6**, **Email** or **Users**.
3. Make the changes. If the tab has a **Save** button, click it.
4. Click the X at the top right to close **Settings**.

Use the moon icon on the top bar to change between the light theme and the dark theme.

Related chapters: 2 Installation, 4 Projects and files.

4 Projects and files

This chapter tells you how to open, import, save and export projects, and where Stratum 4D keeps them.

Projects and files

In this manual, a **project** is a schedule that Stratum 4D keeps in its database. A **file** is a file on the disk, for example an XER file from P6.

You import a file to make a project. You export a project to make a file. The project database keeps each saved version of a project.

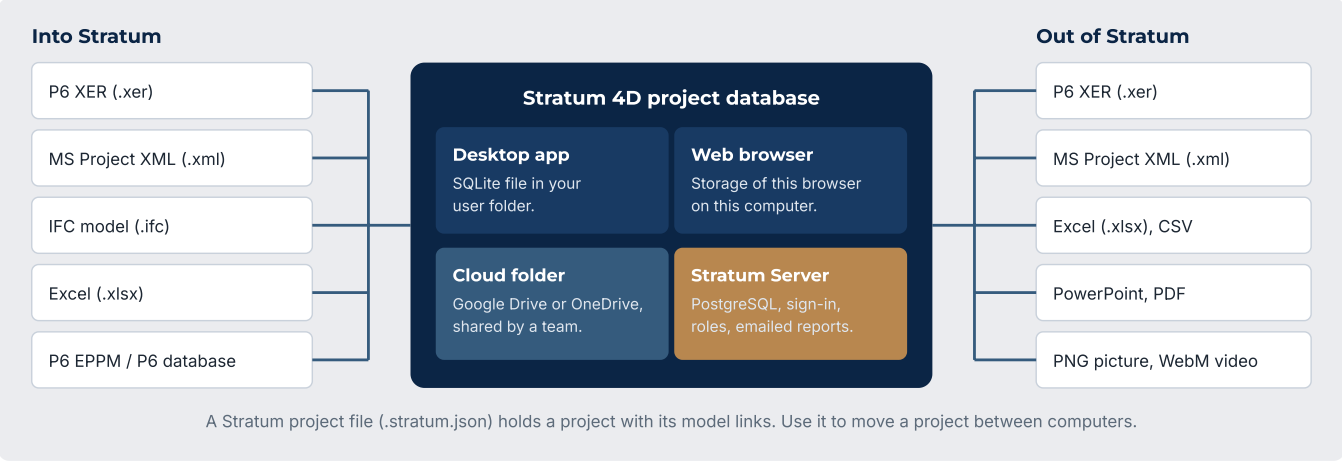


Figure 4-1. Files into and out of Stratum 4D, and the four places for the project database.

Where Stratum 4D keeps the projects

Storage	Where	Who can open the projects
Desktop application	A SQLite file in your user folder	You, on this computer
Web application	The storage of the browser	You, in this browser on this computer
Cloud folder	A Google Drive or OneDrive folder	All persons with edit access to the folder
Stratum Server	A PostgreSQL database	All users of the server, by their role

Each save gets a version number. If a different person saved the project after you opened it, Stratum 4D does not save over that version. It gives you the option **Save as new**.

Import a schedule file

Stratum 4D can import these schedule files:

- Primavera P6 XER files (`.xer`), with activities, WBS, logic, codes, user fields, calendars, resources, costs and baselines.
- Microsoft Project XML files (`.xml`). In Microsoft Project, use **File › Save As › XML Format** to make this file.
- Microsoft Excel files (`.xlsx`) that you exported from Stratum 4D and changed.

> **NOTE**
> Stratum 4D cannot open Microsoft Project `.mpp` files. Save the file as XML in Microsoft Project first.

To import a schedule in the P6 layout

1. Click **File › Import...**.
2. Select the `.xer` or `.xml` file.
3. Click **Open**.
   Result: The schedule opens as a new project.

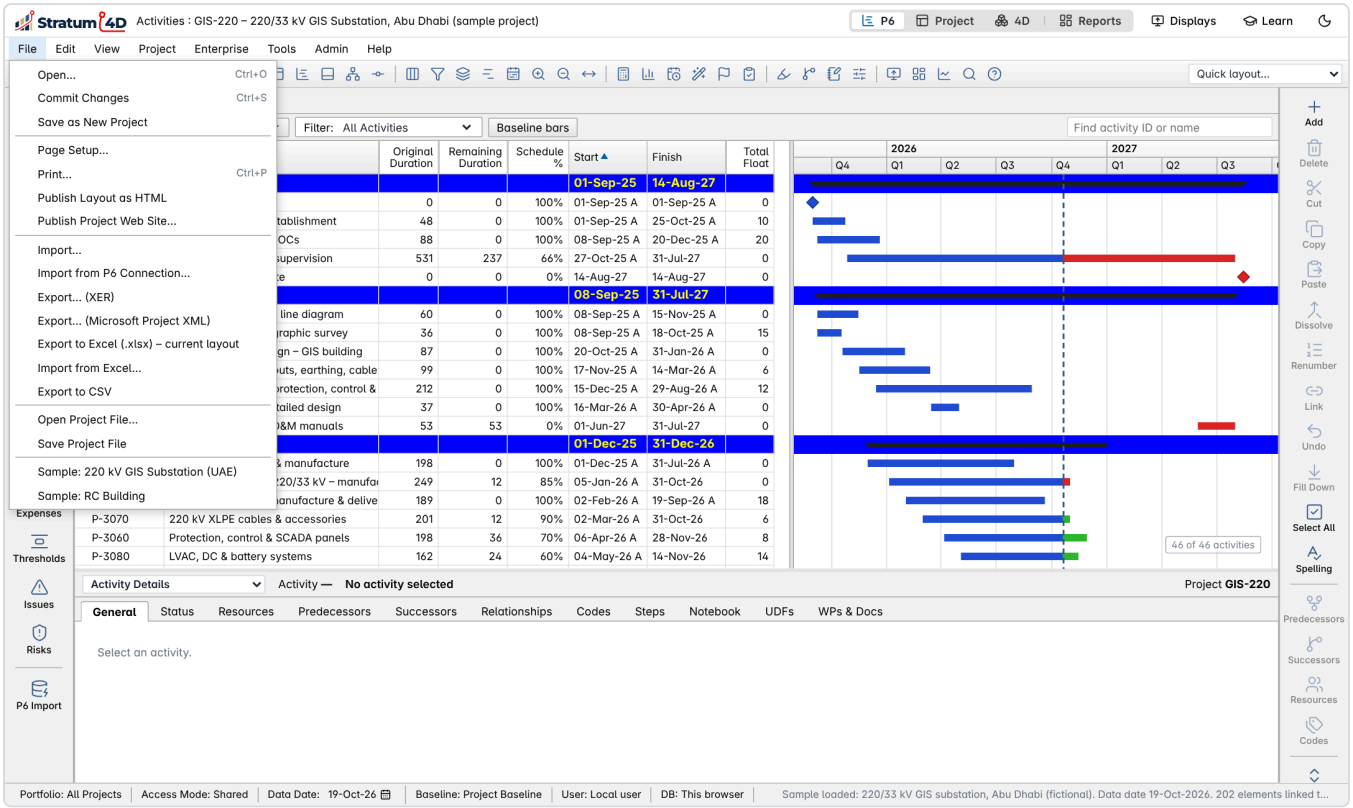


Figure 4-2. The File menu of the P6 layout, with Import and Export.

To import a schedule in the Project layout

1. Click **File**.
2. Click **Open**.
3. Click **Import XER / Microsoft Project XML**.
4. Select the file and click **Open**.

To import a new version of the same schedule

Each week, you can import the new XER over the open project. Stratum 4D keeps the model links because it finds each activity by its Activity ID. If you click **Cancel** at step 3, the file opens as a new project.

1. Open the project.
2. Import the new XER file as usual.
3. Stratum 4D asks "Update the open schedule with this file?". Click **OK**.
   Result: The **Check** tab lists the changes: finish slip, new and deleted activities, and changes of status.

Open and save a project

To open a project from the database

1. Click **Projects** (P6 and 4D layouts), or **File › Open › Projects** (Project layout).
2. Select the project.
3. Click **Open**.

To save a project

1. Push `Ctrl` + `S`, or click **Save**.
   Result: Stratum 4D saves a new version of the project in the database.

Stratum 4D also saves your work in the background every few seconds. If the application closes without a save, it asks you to restore the work at the next start.

Export a project

Export	Use it for
P6 XER	Import into Primavera P6. Use **Update Existing Project** in P6 to bring the dates back.
Microsoft Project XML	Open in Microsoft Project 2010 or later.
Excel (.xlsx) or CSV	Change data in Excel, or make your own reports.
Stratum project file (.stratum.json)	Move a project, with its model links, to a different computer.

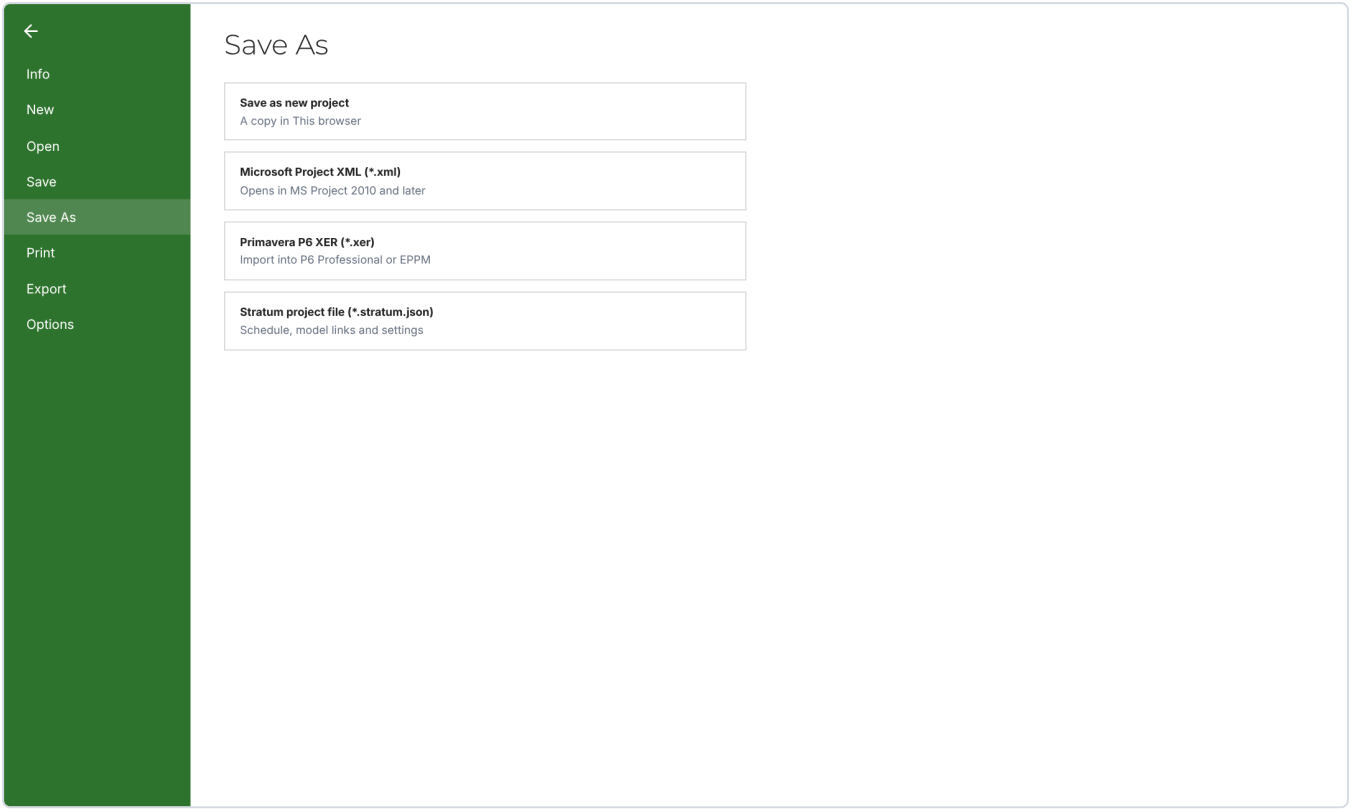


Figure 4-3. File › Save As in the Project layout.

To export a project

1. In the P6 layout, click **File › Export...**. In the Project layout, click **File › Save As**.
2. Click the type of file.
   Result: The file goes into the Downloads folder.

> **CAUTION**
> Before you import an exported XER into P6, make a backup of the P6 project. An import with **Update Existing Project** changes the project in P6.

Share projects with a cloud folder

A team can use one Google Drive or OneDrive folder as a shared database. All persons with edit access to the folder work on the same projects. Stratum 4D does a check before each save, so no person saves over newer work.

1. Click the gear icon (**Settings**).
2. Click the **Data & sign-in** tab.
3. Under **Cloud folder**, click **Google Drive** or **OneDrive**.
4. Sign in to the account.
   Result: A list of your folders opens.
5. Open the folder that your team shares.
6. Click **Use "<folder>" as the Stratum folder**.
   Result: The projects in the folder show in **Projects**.

> **NOTE**
> Your administrator must first register Stratum 4D with Google or Microsoft. The file `docs/CLOUD_STORAGE.md` tells the administrator how to do this.

Related chapters: 3 First start. Later chapters: the P6 layout, the Project layout and the 4D layout.